

AI TRANSFORMATION PROPOSAL

Intelligent Healthcare for Enugu.

A comprehensive AI infrastructure plan for Enugu International Hospital — covering patient intake, pharmacy automation, diagnostics, ward management, billing, security, smart building, and telemedicine.

PREPARED FOR

Enugu International Hospital
Gov. Peter Mbah Initiative

PREPARED BY

Mystra AI
AI Solutions Agency

INVESTMENT RANGE

₦180M – ₦520M
Two-Tier Deployment

Africa’s Most Intelligent Hospital.

Enugu International Hospital has a once-in-a-decade opportunity to become the first fully AI-augmented quaternary hospital in Southeast Nigeria — cutting operational costs, eliminating diagnostic delays, and delivering a patient experience that rivals any world-class facility.

THE CHALLENGE

Nigeria loses ₦180B annually to preventable hospital inefficiencies. Paper records, long queues, pharmacy stockouts, and reactive care hit a 300-bed quaternary facility especially hard. EIH cannot afford to operate manually at this scale.

THE SOLUTION

Nine AI pillars covering every touchpoint: AI triage, appointment scheduling, diagnostics support, ward management IoT, pharmacy robot, billing AI, security AI, smart building management, and telemedicine platform.

TIER 1 — CORE (₦180–250M)

AI Intake & Triage, Appointment Scheduling, CDSS, Smart Ward IoT (300 beds), Pharmacy Automation — live in 6 months. Measurable ROI within 90 days of go-live.

TIER 2 — FULL STACK (₦200–270M)

Billing AI, AI Security, Smart Building BMS/HVAC, Telemedicine Platform — complete AI hospital in 12 months. EIH becomes West Africa’s healthcare technology benchmark.

300–500%

ROI on
scheduling AI

90%

Reduction in
pharmacy errors

25–35%

Smart building
energy savings

THE OPPORTUNITY

Nigeria Spends #1B+ on Medical Tourism Yearly.

ElH's founding mission is to stop the bleeding — to give Southeast Nigerians world-class care without the Emirates flight. But a 300-bed quaternary facility running on manual workflows defeats that purpose from day one. AI is not optional at this scale. It is the infrastructure.

47 min

Average Nigerian outpatient wait time. AI triage cuts this to under 8 minutes.

68%

Of in-hospital medication errors traced to manual pharmacy workflows in Nigeria.

#180B

Lost annually to preventable Nigerian hospital inefficiencies.

\$39.25B

Global AI healthcare market today. \$504B by 2032. ElH moves now or follows.

"African hospitals that adopt AI infrastructure in 2025–2027 will hold a structural cost advantage that compounds for a decade." — McKinsey Global Health, 2024

Complete Hospital Intelligence

Every department. Every workflow. Every outcome.

- 01

AI PATIENT INTAKE & TRIAGE

Self-service kiosks + ESI scoring voice bot (English/Igbo/Pidgin). 83% wait time reduction. 75.7% AI triage accuracy (PMC peer-reviewed).
- 02

APPOINTMENT SCHEDULING AI

WhatsApp + web booking, ElevenLabs voice reminders, AI no-show prediction + auto-fill. 300–500% ROI (Medozai/IntuitionLabs 2025).
- 03

AI DIAGNOSTICS SUPPORT (CDSS)

Aidoc/Google Health CXR + PACS integration. Evidence-based treatment prompts, drug interaction alerts. 40% diagnostic accuracy improvement.
- 04

WARD MANAGEMENT & NURSE CALLS (IoT)

IoT bed sensors, RTLS patient tracking, smart nurse panels, continuous vitals + AI Early Warning Scoring. 25% fewer ICU transfers.
- 05

PHARMACY AUTOMATION (OMNICELL XR2)

Omnice ll XR2 robot + 10 ADC cabinets. 500+ Rx/day. 90% dispensing error reduction. AI inventory reorder (Omnice ll verified).
- 06

BILLING & INSURANCE AI

NLP ICD-10/11 coding, NHIS integration, auto claims. 60% claim denial reduction. Revenue cycle 40% faster (45 days → under 14).
- 07

AI SECURITY & CCTV

Hikvision DeepinMind 4K × 120 cameras. HID biometric access × 40 points. Real-time AI threat detection and anomaly alerts.
- 08

SMART BUILDING (BMS / HVAC)

Siemens Desigo CC or Honeywell EBI. AI-driven HVAC, power, lighting. 25–35% energy cost reduction (HealthTech Magazine 2025).
- 09

TELEMEDICINE PLATFORM

React Native app (iOS + Android), Daily.co/Vonage video. CommonSpirit command-centre model. Rural Enugu State coverage.

No More 47-Minute Queues.

CURRENT STATE — PROBLEMS

- Patients queue 45–90 min before seeing triage nurse
- Paper registration creates errors and lost records
- No-shows waste 30–40% of outpatient capacity
- Critical patients wait behind routine cases
- Staff spend 40% of time on admin, not clinical care
- No data on demand, peak times, or patient flow

MYSTRA AI SOLUTION

- AI voice triage bot (English, Igbo, Pidgin) pre-screens before arrival
- Digital kiosk registration linked to EHR from first contact
- Smart scheduling with AI no-show prediction + auto-fill
- ESI-based acuity routing — severity determines queue
- WhatsApp + SMS automated appointment reminders
- Real-time demand forecasting for staffing and resources

83%

Reduction in
average wait time

75.7%

AI ESI triage
accuracy (PMC)

40%

More patients served
per day, same staff

02

PILLAR 02 — APPOINTMENT SCHEDULING AI

Scheduling That Books Itself.

Patients book via WhatsApp, web, or EIH’s mobile app. AI predicts no-shows and auto-fills cancelled slots. An ElevenLabs voice agent calls patients 24h and 2h before appointments in their preferred language. The system learns from EIH’s booking history to optimise physician utilisation and reduce idle capacity from ~35% to under 10%.

300–500%
Net ROI on
scheduling AI
(Medozai 2025)

03

PILLAR 03 — AI DIAGNOSTICS SUPPORT (CDSS)

A Consultant That Never Sleeps.

Clinical Decision Support integrates with EIH’s EHR (Epic or Oracle Health). It surfaces evidence-based treatment recommendations, flags abnormal labs, catches drug-drug interactions, and generates structured radiology reports via Aidoc/Google Health CXR. Built on Nigerian clinical data + WHO/NICE guidelines. Cost per consultation after deployment: ₦0.

40%
Improvement in
diagnostic accuracy

Wards That Think. Emergencies That Never Surprise.

IOT MONITORING NETWORK

Smart Ward Infrastructure

- Continuous vitals per bed: SpO2, BP, HR, Temp, RR
- RTLS sensors tracking patient location in real time
- Early Warning Score calculated every 15 minutes
- Smart nurse panels with AI-prioritised alert routing
- Fall prevention: movement pattern + bed exit alerts
- Ventilator and IV pump anomaly detection

25%
Fewer preventable ICU transfers

PREDICTIVE ENGINE

Sepsis & Deterioration Early Warning

- Aggregates vitals, labs, imaging, notes in real time
- Sepsis detection 6–12 hours before clinical symptoms
- AKI and Cardiac deterioration models built in
- Predictive readmission scoring — flags risky discharges
- Risk stratification dashboard for ward rounds
- Auto-links to CDSS for treatment recommendations

6–12h
Earlier sepsis detection

"At UCH Ibadan, IoT-enabled ward monitoring reduced unexpected ICU admissions by 22% in the first 8 months." — Clinical AI Africa, 2024

05

PILLAR 05 — PHARMACY AUTOMATION

Omnicell XR2. Robotic. Zero Errors.

Omnicell XR2 robotic dispensing handles 500+ daily prescriptions. Ten ADC cabinets distributed across EIH wards ensure nurses access medications at point of care. AI inventory management maintains optimal stock, flags expiry, and triggers auto-reorder. Every prescription verified against the patient’s drug interaction profile before dispensing. Pharmacist time redirected to clinical consultations — not counting pills.

90% Dispensing error reduction verified by Omnicell

06

PILLAR 06 — BILLING & INSURANCE AI

Revenue That Doesn’t Leak.

NLP-powered ICD-10/11 coding converts clinical notes to billing codes automatically. Claims submitted to NHS and private insurers with 97% first-pass accuracy. Revenue cycle drops from 45 days to under 14. The system identifies undercoded visits and missed charges — recovering revenue that currently evaporates through manual processes.

60% Reduction in claim denials

A Hospital That Protects Itself Automatically.

PILLAR 07 — AI SECURITY

Hikvision DeepInMind AI Security Network

- 120 × 4K DeepInMind cameras with real-time threat detection
- 40 × HID biometric access control points
- Perimeter intrusion detection + automatic alert escalation
- Loitering and abandoned-object detection in public areas
- Facial recognition for VIP patient and staff access
- Central security ops centre with live AI dashboard

120
AI cameras across EIH campus

PILLAR 08 — SMART BUILDING

Siemens Desigo CC BMS / HVAC Control

- Unified BMS for HVAC, power, lighting, and lifts
- AI occupancy-based climate control per ward
- Predictive maintenance — flags faults before failure
- Medical gas monitoring with automatic leak detection
- Generator auto-switch with AI load balancing
- Real-time energy dashboard for facilities management

25–35%
Energy cost reduction

ElH's Reach. All of Enugu State.

Modelled on CommonSpirit Health's 30-command-centre telemedicine network, ElH's platform connects rural communities across Enugu State to specialist consultations without requiring physical travel. Built on React Native for iOS and Android, powered by Daily.co or Vonage video infrastructure.

VIDEO CONSULTATION

HD Video + Medical Peripherals

4 on-site telemedicine booths at ElH with digital stethoscopes, dermatoscopes, and BP monitors. Remote clinicians see and hear everything.

PATIENT APP

iOS + Android App

ElH branded app: book appointments, view results, attend video consultations, receive medication reminders, message care teams.

RURAL REACH

Community Health Worker Integration

CHWs in rural Enugu equipped with tablets act as local endpoints — ElH specialists consult on complex cases remotely, no travel required.

AI PRE-SCREENING

Symptom Check Before Every Call

AI screens patient symptoms before every virtual appointment and flags urgent cases for immediate escalation to in-person emergency care.

ElH's telemedicine platform positions the hospital as the healthcare hub for all of Enugu State — not just patients who can physically reach Rangers Avenue.

+35%

RETURN ON INVESTMENT

Where the Numbers Land.

All projections sourced from published African hospital AI deployments — UCH Ibadan, Kenyatta National Hospital, Aga Khan Nairobi — adjusted for EIH's 300-bed capacity. Conservative estimates only.



Based on comparable African hospital deployments, EIH's full-system investment pays back within **24-36 months** of full Tier 2 go-live. After that, AI operations cost less than 2% of total hospital revenue annually.

24mo

What Gets Built.

Tier 1 hardware procured through validated healthcare technology suppliers. All AI inference runs on-premise — patient data stays in Nigeria, NDPA-compliant by design, fully operational without internet dependency.

Component	Specification	Qty	Est. Cost (₦)
Omnicell XR2 Robotic Pharmacy Unit	500+ Rx/day automated dispensing + Nigerian import/install	1	₦38.9M
Automated Dispensing Cabinets (ADC)	Point-of-care medication access distributed across all wards	10	₦22.4M
IoT Vital Signs Sensors (per bed)	Wireless: SpO2, BP, HR, Temperature, Respiratory Rate	300	₦24.6M
AI Server Infrastructure	On-premise GPU inference (NVIDIA A100 class) + NAS storage	2	₦18.5M
RTLS Infrastructure	Real-time location for patients, staff, and equipment	1 lot	₦8.4M
Patient Registration Kiosks	Touchscreen self-service with fingerprint + camera	8	₦7.8M
Network Backbone Upgrade	Hospital-wide fiber, Wi-Fi 6E APs, medical device network segmentation	1 lot	₦12.2M
Software Licences & EHR Integration	CDSS, Scheduling AI, Admin AI, EHR connect — 2yr SLA	1 lot	₦27.6M
Tier 1 Total			₦160M

* Tier 2 (Security, BMS/HVAC, Billing AI, Telemedicine) adds ₦180–₦360M billed at Tier 2 sign-off. Final figures subject to USD/NGN at procurement date.

AFRICA PRECEDENTS

It Has Already Been Done **Here.**

- 01

University College Hospital (UCH), Ibadan — NIGERIA

WhatsApp telemedicine and AI-assisted patient flow deployed post-2021. Specialist utilisation increased by 35%. UCH is now cited in Nigeria's 2025 National Digital Health Policy as proof of AI feasibility in Nigerian public tertiary hospitals.

+35% specialist utilisation increase
- 02

Kenyatta National Hospital — KENYA

Google Health AI radiology deployed 2022. TB detection rates improved significantly. Radiologist reporting time fell from 4.2 hours to 38 minutes. Now mandated across 14 Kenyan public hospitals by the Ministry of Health — cited in Kenya's 2025 National AI Strategy.

3.3x faster radiology reporting — 4.2h to 38 minutes
- 03

Aga Khan University Hospital, Nairobi — KENYA / TANZANIA / UGANDA

Epic EHR + Honeywell BMS deployed across the network. Nairobi campus (500+ beds) achieved 28% energy reduction. AI billing recovered \$2.1M USD in undercoded revenue in year one. Full-suite AI deployment since 2021 is Aga Khan's operational standard across East Africa.

28% energy reduction + \$2.1M recovered billing revenue

TOTAL INVESTMENT RANGE

Full 9-pillar AI hospital transformation, delivered in two tiers over 12 months

#520M

Maximum. Minimum starts at #180M for Tier 1 alone.

TIER 1 — CORE FOUNDATION

AI Triage, Pharmacy,
Ward & Diagnostics

#180–#250M

Includes: AI Triage + Registration Kiosks · Appointment Scheduling AI (ElevenLabs) ·
CDSS + EHR Integration · IoT Ward Monitoring 300 beds · Omnicell XR2 + 10 ADC cabinets
· Network backbone · All hardware, install + 12-month SLA (#18–24M/year)

Payback: 18–24 months

RECOMMENDED — FULL STACK

TIER 2 — FULL INTELLIGENCE

Security, BMS, Billing
& Telemedicine

#200–#270M

Includes: Hikvision DeepinMind × 120 + HID × 40 · Siemens Desigo CC BMS/HVAC · ICD-
10/11 Billing AI + NHIS · React Native Telemedicine App · Executive Intelligence Dashboard
· Staff AI Training Academy · Extended 12-month SLA

Est. #80–120M/year savings post-payback

PAYMENT SCHEDULE — PER TIER

30%

On contract signing

40%

At hardware delivery + install

30%

At go-live sign-off

BEGIN THE TRANSFORMATION

The Future of Healthcare Starts Here.

Schedule a 60-minute executive presentation with the Mystra AI hospital team. We will walk EIH leadership through the full technical architecture, introduce our implementation team, and answer every question before a single naira is committed.

EMAIL

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WEBSITE

mystrahq.com

NEXT STEP

Book Executive Briefing
No obligation. 60 minutes.